

6.2 - Hamilton Paths and Circuits

Problem Solving, Math 1000

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Question

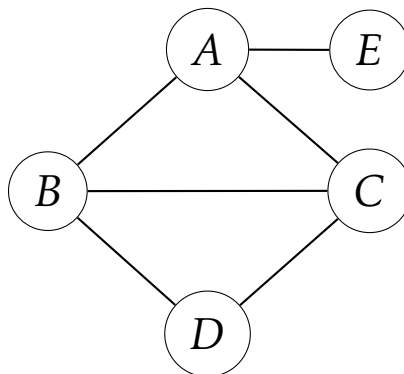
Previously, we discussed how to find a path that traversed every edge of a graph once and only once. What if we were to ask for a path that traversed each vertex once and only once?

Definition

A **Hamilton path** in a connected graph is a path that visits each vertex once and only once. A **Hamilton circuit** in a connected graph is a circuit that visits each vertex once and only once.

Example

The following graph has a Hamilton path given by E, A, B, D, C (there are others, but this is just one example):



Theorem

Any graph which has a Hamilton circuit has a Hamilton path, which may be formed by deleting the final edge in the circuit.